Notes: Laibson (QJE, 1997)

Golden Eggs and Hyperbolic Discounting

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1 Big picture

- **Research question.** How do dynamically inconsistent consumers choose saving and consumption when they can partially commit their future selves through illiquid assets?
- **Main idea.** Hyperbolic discounting creates a conflict between today's patient self and tomorrow's impatient self. Illiquid assets, such as housing equity, pensions, savings bonds, certificates of deposit, or durable goods, act like "golden eggs": they preserve long-run value but are costly or slow to turn into current consumption.
- **Commitment technology.** The consumer can hold a liquid asset x and an illiquid asset z. Liquid wealth can be consumed immediately, while selling or borrowing against z requires a one-period delay. This delay prevents the current self from instantly consuming all wealth.
- **Core mechanism.** Earlier selves deliberately put wealth into illiquid form. Later selves then face self-imposed liquidity constraints and consume mainly out of current cash flow.
- **Main predictions.** The model predicts consumption-income tracking, asset-specific marginal propensities to consume, violations of Ricardian equivalence, and a possible fall in saving when financial innovation makes illiquid assets easier to borrow against.
- **Key welfare message.** More liquidity is not always better. When consumers value commitment, financial innovation that removes illiquidity can lower welfare by weakening self-control.
- **Economic takeaway.** The paper provides a disciplined model in which apparently behavioral phenomena–mental accounts, excess sensitivity of consumption, and demand for illiquid assets–arise from sophisticated agents who understand their own future self-control problems.

2 Incremental contribution of the paper

- **From Strotz-style commitment to macro-finance.** Earlier work had shown that nonexponential discounting creates demand for commitment. Laibson embeds that idea in a tractable consumption-saving model with assets resembling common household balance-sheet items.
- **Quasi-hyperbolic preferences.** The paper popularizes the discrete-time β-δ formulation,

$$U_t = E_t\left[u(c_t) + \beta \sum_{\tau=t+1}^{T} \delta^{\tau-t} u(c_\tau)\right],$$

  which captures present bias while preserving much of the tractability of exponential discounting.
- **Illiquidity as commitment.** Instead of assuming an abstract commitment device, the model uses an imperfect but economically meaningful technology: a delayed-access asset.
- **Intrapersonal game.** The consumer is modeled as a sequence of temporal selves. Equilibrium is a subgame-perfect equilibrium of this dynamic game.
- **Microfoundation for excess sensitivity.** Consumption tracks predictable income not because consumers are naive or liquidity constrained by banks, but because earlier selves intentionally restrict later selves' liquidity.

- **Asset-specific MPCs without mechanical mental accounting.** The model predicts a high marginal propensity to consume out of liquid income and a low marginal propensity to consume out of illiquid wealth.
- **Financial innovation can be harmful.** The paper makes a then-unusual point: improving household access to credit may reduce saving and welfare if it destroys valuable commitment opportunities.

3 Model setup and measurement

3.1 Preferences

The consumer has present-biased, dynamically inconsistent preferences. At date t, the current self gives normal exponential weight to all future tradeoffs after tomorrow, but applies an extra discount factor $\beta < 1$ to all utility after the current period:

$$U_t = E_t \left[u(c_t) + \beta\delta u(c_{t+1}) + \beta\delta^2 u(c_{t+2}) + \cdots \right].$$

The instantaneous utility function is CRRA, with curvature parameter ρ. When $\beta = 1$, the model collapses to the standard exponential-discounting benchmark. When $\beta < 1$, the consumer has present bias: future selves will want to consume more immediately than the current self would like them to consume.

Interpretation. The agent is not myopic. She foresees her future preference reversals and tries to shape future choice sets.

3.2 Assets and timing

Each period has two asset accounts:

- **Liquid asset** x**.** Liquid wealth can be consumed immediately.
- **Illiquid asset** z**.** Sales or collateralized borrowing against this asset must be initiated one period before proceeds can be consumed.

The current consumption constraint is

$$c_t \leq y_t + R_t x_{t-1},$$

while end-of-period asset allocation satisfies

$$x_t + z_t = R_t(x_{t-1} + z_{t-1}) + y_t - c_t, \qquad x_t, z_t \geq 0.$$

The nonnegativity constraints are important. They rule out fully flexible forced-saving contracts that would let today's self perfectly commit tomorrow's self.

Interpretation. The one-period delay is the model's golden-eggs feature. The asset preserves long-run wealth, but the current self cannot crack it open immediately.

3.3 Intrapersonal game

A finite-horizon consumption problem becomes a game among selves $1, 2, \ldots, T$. Self t controls period-t consumption and asset allocation. The equilibrium concept is subgame perfection.

The paper imposes a restriction on deterministic labor-income paths:

$$u'(y_t) \geq \beta\delta^{\tau} \left(\prod_{i=1}^{\tau} R_{t+i} \right) u'(y_{t+\tau}) \quad \forall t, \tau \geq 1.$$

This condition keeps income fluctuations within a band large enough for realistic variation but small enough to deliver a clean equilibrium characterization.

3.4 Equilibrium characterization

The main theorem establishes a unique resource-exhausting subgame-perfect equilibrium under the income-path restriction. Equilibrium strategies satisfy four intuitive properties:

- marginal utility cannot be too low relative to future marginal utility, because the self can always save;
- if marginal utility is strictly too high, the current liquidity constraint binds;
- if tomorrow's self would otherwise overconsume, today's self sets tomorrow's liquid assets as low as possible;
- if tomorrow's self would otherwise underconsume, today's self leaves more wealth liquid.

Interpretation. The equilibrium is not a standard Euler-equation allocation. It is a strategic allocation of liquidity across selves.

4 Main mechanisms and empirical implications

4.1 Consumption-income tracking

The model explains why consumption can track predictable labor income even when consumers own substantial assets. The reason is that most assets are deliberately held in illiquid form. When cash income is high, the current self consumes more; when cash income is low, the current self is constrained and consumes less.

This differs from a standard liquidity-constraint story. The constraint is endogenous and strategic, not necessarily imposed by outside lenders.

4.2 General-equilibrium steady state

Laibson embeds the consumption problem in a simple production economy. Under the golden-eggs technology, the steady state satisfies

$$\exp(\rho g) = R\delta,$$

where g is the steady-state growth rate and R is the gross return. A striking feature is that β does not enter this steady-state relationship when commitment is available.

Economic intuition. Present bias affects short-run liquidity management, but the illiquid asset allows earlier selves to discipline later selves. Long-run capital accumulation can therefore be pinned down by the standard long-run discounting parameter δ rather than by the short-run temptation parameter β.

4.3 Asset-specific marginal propensities to consume

On the equilibrium path, the marginal propensity to consume out of liquid assets is one:

$$\frac{\partial c_t}{\partial(R_t x_{t-1})} = 1.$$

By contrast, the marginal propensity to consume out of illiquid assets is zero at the current date:

$$\frac{\partial c_t}{\partial(R_t z_{t-1})} = 0.$$

The paper also studies a longer-run geometric-average measure of the marginal propensity to consume out of illiquid wealth, which is close to zero for reasonable parameters.

Economic intuition. Cash in hand is tempting and immediately consumed. Illiquid wealth is protected from the current self and therefore behaves like a separate mental account.

4.4 Ricardian equivalence

The timing of cash flows matters. A tax cut today financed by a tax increase tomorrow changes current liquidity even if lifetime resources are unchanged. Because current selves are liquidity constrained by earlier selves' asset choices, present-value equivalence does not imply behavioral equivalence.

Implication. Ricardian equivalence fails even for a representative, infinitely lived household with substantial assets.

4.5 Financial innovation and declining saving

The paper argues that the expansion of credit cards, ATM networks, and home-equity lines in the 1980s made it easier for households to borrow against formerly illiquid wealth. If consumers can instantaneously borrow against z, the illiquid asset no longer provides commitment.

In the instantaneous-credit economy, the steady state satisfies

$$\exp(\rho g) = \beta\delta R + (1-\beta)\delta\exp(g).$$

For $\beta < 1$, this implies lower capital accumulation than in the golden-eggs economy.

Economic intuition. Credit-market innovation relaxes the exact constraint that made illiquid assets useful. This can raise current consumption, reduce saving, and lower steady-state capital.

4.6 Welfare analysis

The model compares a golden-eggs economy with an instantaneous-credit economy. For reasonable β values, the representative consumer must be paid a positive amount to be indifferent between keeping the golden-eggs technology and switching to the instantaneous-credit technology.

Interpretation. Extra liquidity expands the feasible set of the current self, but it can shrink the commitment set available to earlier selves. For a sophisticated hyperbolic consumer, that loss of commitment can dominate.

5 Identification and interpretation

5.1 What the paper can identify

This is primarily a theoretical and calibration paper. It does not estimate causal effects using microdata. Instead, it shows that a small set of behavioral and institutional assumptions can jointly rationalize several observed facts about household consumption and saving.

5.2 Why the quasi-hyperbolic form matters

The β-δ formulation separates short-run impatience from long-run patience. This separation is essential because the model can match both:

- high short-run sensitivity of consumption to cash flow through $\beta < 1$; and
- plausible long-run capital accumulation through δ.

5.3 Role of sophistication

The consumer understands future self-control problems. The demand for illiquid assets is therefore not a mistake. It is a rational response to future temptation.

5.4 Interpretation of illiquidity

Illiquidity is not only a transaction cost or market imperfection. In this model it is also a service: it helps consumers protect wealth from future selves.

5.5 Limits of the calibration evidence

The paper's calibration of the saving decline is illustrative. Other forces also affected U.S. saving in the 1980s and 1990s, and the model abstracts from uncertainty, heterogeneous households, tax incentives, bequests, default, and precautionary liquidity motives.

5.6 Interpretation of financial innovation

The paper does not claim that credit access is always bad. Liquidity can help households smooth shocks and handle emergencies. The narrower claim is that, when self-control is important, some forms of liquidity can destroy valuable commitment.

6 Figures and tables

6.1 Figure I: Discount functions

Read:

- Figure I compares exponential, generalized hyperbolic, and quasi-hyperbolic discount functions.
- The exponential curve declines smoothly with a constant discount rate.
- The hyperbolic and quasi-hyperbolic curves place especially low weight on near-future utility relative to current utility.
- The quasi-hyperbolic function creates a discrete present-bias wedge: the discount factor drops sharply after the current period and then declines approximately exponentially.

Economic message: The picture shows why time inconsistency arises. From today's perspective, future tradeoffs look relatively patient; when a future date becomes the present, immediate consumption receives a special premium.

6.2 Figure II: Consumption and labor income

Read:

- Figure II plots a deterministic labor-income path and the corresponding equilibrium consumption path.
- Consumption follows predictable movements in labor income rather than smoothing them away.
- The consumption line is less jagged than labor income but still clearly comoves with it.
- This happens even though the consumer has assets, because earlier selves keep wealth illiquid to restrain later selves.

Economic message: The central consumption implication is excess sensitivity to income. In this model, income tracking reflects self-control rather than irrationality or external borrowing constraints.

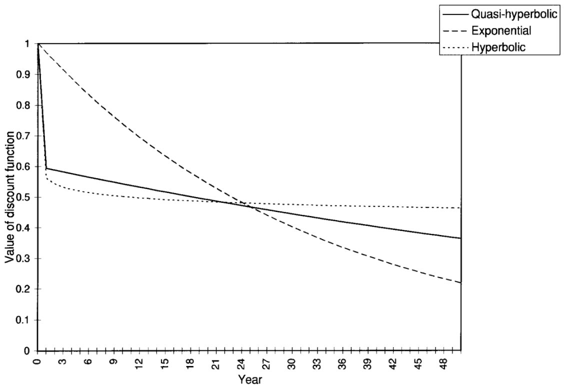

FIGURE I
Discount Functions

Figure 1: Discount functions.

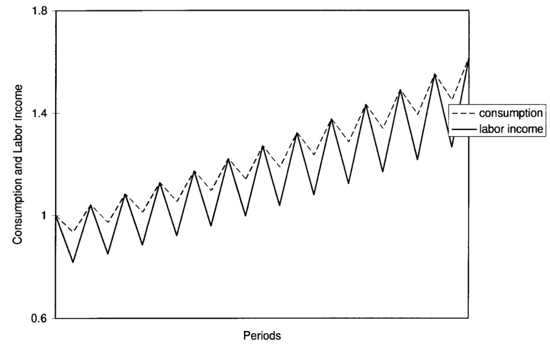

FIGURE II
Consumption and Labor Income

Figure 2: Consumption and labor income.

6.3 Table I: Steady-state interest rates and capital-output ratios

Read:

- Table I compares economies with partial commitment and without partial commitment.
- With commitment, the interest rate is 4.0 percent and the capital-output ratio is 3.00 for all listed values of β.
- Without commitment, lower β values imply higher interest rates and lower capital-output ratios.
- For $\beta = 0.6$, the interest rate rises to 5.3 percent and the capital-output ratio falls to 2.70.
- When $\beta = 1$, the consumer is dynamically consistent, and the two economies coincide.

Economic message: The table quantifies the saving effect of lost commitment. Present bias matters for aggregate capital only when financial technology lets consumers defeat their own commitment devices.

6.4 Table II: Welfare cost of instantaneous credit

Read:

- Table II reports the payment required to make the consumer indifferent between the golden-eggs economy and the instantaneous-credit economy.
- Positive values mean the consumer would need compensation to accept the no-commitment financial technology.
- The payment is very large when present bias is strong: 69.6 percent of output for $\beta = 0.2$.
- For $\beta = 0.6$, the compensation is 9.0 percent of output.
- For $\beta = 1$, the welfare cost is zero because the consumer no longer needs commitment.

Economic message: Liquidity has a shadow cost when it undermines commitment. Financial innovation can be privately tempting but welfare reducing for a sophisticated present-biased consumer.

TABLE I
STEADY STATE INTEREST RATES AND CAPITAL-OUTPUT RATIOS IN ECONOMIES WITH AND WITHOUT PARTIAL COMMITMENT

	With commitment (i.e., no instantaneous credit)		Without commitment (i.e., instantaneous credit)	
	r	$\frac{K}{Y}$	r	$\frac{K}{Y}$
$\beta = 0.2$	0.040	3.00	0.119	1.81
$\beta = 0.4$	0.040	3.00	0.070	2.40
$\beta = 0.6$	0.040	3.00	0.053	2.70
$\beta = 0.8$	0.040	3.00	0.045	2.88
$\beta = 1.0$	0.040	3.00	0.040	3.00

Figure 3: Steady-state interest rates and capital-output ratios.

TABLE II
PAYMENTS TO INDUCE INDIFFERENCE BETWEEN GOLDEN EGGS ECONOMY AND INSTANTANEOUS CREDIT ECONOMY

	Payment as percent of output
$\beta = 0.2$	69.6
$\beta = 0.4$	29.5
$\beta = 0.6$	9.0
$\beta = 0.8$	1.6
$\beta = 1.0$	0.0

Figure 4: Payments to induce indifference.

7 Short summary

- The paper studies a sophisticated hyperbolic consumer who values commitment.
- Preferences are quasi-hyperbolic: $\beta < 1$ captures present bias and δ captures long-run patience.
- The consumer holds liquid assets and illiquid assets.
- Illiquid assets are useful because they cannot be consumed immediately.
- Earlier selves use illiquidity to restrain later selves.
- The equilibrium features self-imposed liquidity constraints.
- Consumption tracks predictable income because cash flow is consumed when it arrives.
- The marginal propensity to consume is high out of liquid wealth and low out of illiquid wealth.
- The model violates Ricardian equivalence because the timing of cash flows matters.
- Financial innovation that enables instantaneous borrowing against illiquid assets weakens commitment.
- The model predicts lower saving and lower capital accumulation after such innovation.
- For reasonable parameter values, consumers may be worse off with too much liquidity.

8 Conclusion

8.1 Core conclusion

Laibson shows that illiquid assets can be valuable commitment devices for consumers with hyperbolic discounting. A consumer who understands her own future present bias may intentionally place wealth in assets that are difficult to liquidate. This creates self-imposed liquidity constraints and changes consumption behavior in ways that resemble several empirical puzzles.

8.2 Economic intuition

The consumer is a sequence of selves. The current self wants future selves to save, but each future self will want to consume immediately when its period arrives. Illiquid assets solve part of this problem by slowing access to wealth. The golden-eggs asset keeps producing future benefits but cannot be fully consumed today.

8.3 Incremental contribution revisited

The paper connects behavioral discounting, household finance, and macroeconomic saving. It shows that consumption-income comovement, mental-account-like behavior, and the saving effects of consumer credit can arise from one coherent model of sophisticated present-biased agents.

8.4 Implications

The findings imply that financial regulation and product design should consider the commitment value of illiquidity. A retirement account, pension, mortgage, or savings bond may be valuable not only because of returns or tax treatment, but because it limits future temptation. Conversely, credit cards and home-equity borrowing can reduce commitment by making long-term wealth spendable today.

8.5 Limitations

The model is deliberately stylized. It uses deterministic income in the main analysis, a simple one-period delay technology, and a representative-agent calibration. It also predicts that consumers are always liquidity constrained on the equilibrium path, a feature that later work can soften by adding precautionary motives, uncertainty, richer assets, or heterogeneity.

8.6 Takeaway

The enduring takeaway is that liquidity has two sides. It helps consumption smoothing in standard models, but it can undermine self-control in present-biased models. The welfare value of a financial asset therefore depends not only on return and risk, but also on how it shapes future choice sets.